

## Numerical Techniques 2026–2027

### 2. The oscillation equation and time differencing

Daan Degrauwe

`daan.degrauwe@meteo.be`

Postgraduate Studies in Weather and Climate Modeling

Ghent University

- Positioning of the course in the postgraduate program
- Discretization and finite differences
- Consistency, convergence and stability
- Von Neumann analysis and Courant number  $\mu = c \frac{\Delta t}{\Delta x}$ , linking spatial resolution to temporal resolution
- Example of upstream scheme for 1D advection equation
- Don't worry about math too much!

- The oscillation equation
- Errors due to time discretization: amplitude and phase
- Concrete schemes
  - ▶ Single stage, two timelevel  
Forward, backward, trapezium
  - ▶ Multistage, two timelevel  
Heun, Matsuno, Runge-Kutta
  - ▶ Single stage, three timelevel  
Leapfrog, Adams-Bashforth
  - ▶ Filter for computational modes

The oscillation equation is given by:

$$\frac{d\psi}{dt} = i\kappa\psi$$

Although much simpler than the partial differential equations (PDE's) in atmospheric modeling, the oscillation equation is very relevant:

- 1 Wave-dominated hyperbolic PDE's can be decoupled in advection equations
- 2 The advection equation in Fourier expansion looks like the oscillation equation

# From advection equation to oscillation equation

- The advection equation

$$\frac{\partial \psi}{\partial t} + c \frac{\partial \psi}{\partial x} = 0$$

in Fourier expansion  $\left( \psi(x, t) = \sum_{k=-\infty}^{\infty} \hat{\psi}_k(t) e^{ikx} \right)$  looks like

$$\frac{d\hat{\psi}_k}{dt} = -ikc \hat{\psi}_k$$

which is actually the oscillation equation

$$\frac{d\hat{\psi}}{dt} = i\kappa \hat{\psi}$$

- The influence of the space discretization is removed by the Fourier transform, so we can now focus on the *time differencing*.

## The oscillation equation: time evolution

- The exact time evolution of the solution is

$$\psi(t_0 + \Delta t) = e^{i\kappa\Delta t}\psi(t_0) \equiv A_e\psi(t_0)$$

- For an approximate (numerical) solution,  $\phi^{n+1} = A\phi^n$ .
- The quality of the approximation will depend on the difference between  $A$  and  $A_e$ .
- Note: we will use the exponential notation of a complex number:

$$A = |A|e^{i\theta}$$

with

$$|A| = \sqrt{\mathcal{R}\{A\}^2 + \mathcal{I}\{A\}^2}$$
$$\theta = \arg(A) = \arctan\left(\frac{\mathcal{I}\{A\}}{\mathcal{R}\{A\}}\right)$$

## The oscillation equation: time evolution

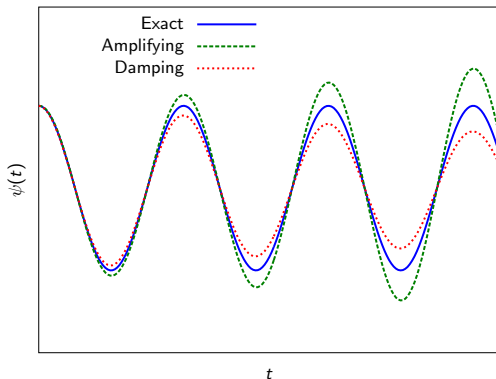
Approximation of exact solution  $A_e = e^{i\kappa\Delta t}$  by numerical solution  $A = |A|e^{i\theta}$ :

- Amplitude

$|A| > 1$       amplifying

$|A| = 1$       neutral

$|A| < 1$       damping



## The oscillation equation: time evolution

Approximation of exact solution  $A_e = e^{i\kappa\Delta t}$  by numerical solution  $A = |A|e^{i\theta}$ :

- Amplitude

$|A| > 1$       amplifying

$|A| = 1$       neutral

$|A| < 1$       damping

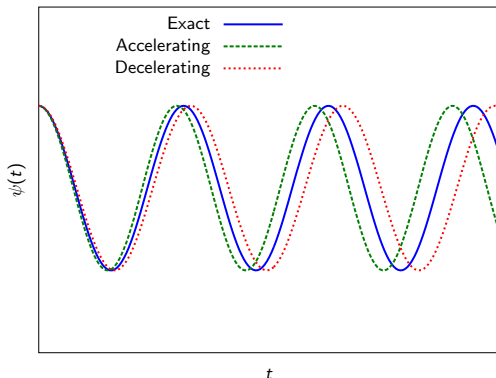
- Phase speed: error is characterized by the *relative* phase change

$$R = \frac{\theta}{\kappa\Delta t}$$

Then

$R > 1$       accelerating

$R < 1$       decelerating





We will now review several schemes:

- Single stage, two-time-level schemes
- Multistage schemes
- Three-time-level schemes

- For the generic equation

$$\frac{d\psi}{dt} = F(\psi)$$

the single-stage two-time-level schemes have the form

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \alpha F(\phi^n) + \beta F(\phi^{n+1})$$

- For the generic equation

$$\frac{d\psi}{dt} = F(\psi)$$

the single-stage two-time-level schemes have the form

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \alpha F(\phi^n) + \beta F(\phi^{n+1})$$

- Some terminology:

|                        |                       |                          |
|------------------------|-----------------------|--------------------------|
| $\alpha = 1$           | $\beta = 0$           | forward differencing     |
| $\alpha = 0$           | $\beta = 1$           | backward differencing    |
| $\alpha = \frac{1}{2}$ | $\beta = \frac{1}{2}$ | trapezoidal differencing |

## Single stage two-time-level schemes: amplification factor

- For the oscillation equation the right-hand side is simply  $F(\psi) = i\kappa\psi$ , so the single-stage two-time-level schemes become

$$(1 - i\beta\kappa\Delta t)\phi^{n+1} = (1 + i\alpha\kappa\Delta t)\phi^n$$

- The amplification factor is then

$$A \equiv \frac{\phi^{n+1}}{\phi^n} = \frac{1 + i\alpha\kappa\Delta t}{1 - i\beta\kappa\Delta t}$$

- The order of accuracy is identified when taking the Taylor expansion of the amplification factor  $A$ :

$$A = 1 + i(\alpha + \beta)\kappa\Delta t - (\alpha + \beta)\beta(\kappa\Delta t)^2 + O(\Delta t^3)$$

while the exact amplification factor is:

$$A_e = e^{i\kappa\Delta t} = 1 + i\kappa\Delta t - \frac{1}{2}(\kappa\Delta t)^2 + O(\Delta t^3)$$

so

- ▶ forward and backward schemes are first order accurate
- ▶ trapezium scheme is second order accurate

- The modulus of the amplification factor is

$$|A|^2 = \frac{1 + \alpha^2 \kappa^2 \Delta t^2}{1 + \beta^2 \kappa^2 \Delta t^2}$$

- So the stability properties are:

- ▶ Forward scheme ( $\alpha = 1, \beta = 0$ )  $\Rightarrow$  unstable (amplifying)
- ▶ Trapezium scheme ( $\alpha = \beta = 1/2$ )  $\Rightarrow$  stable (neutral)
- ▶ Backward scheme ( $\alpha = 0, \beta = 1$ )  $\Rightarrow$  stable (damping)

## Single stage two-time-level schemes: phase error

- The relative phase change is

$$R \equiv \frac{\arg(A)}{\kappa \Delta t} = \frac{1}{\kappa \Delta t} \arctan \left( \frac{(\alpha + \beta) \kappa \Delta t}{1 - \alpha \beta (\kappa \Delta t)^2} \right)$$

- For the forward scheme ( $\alpha = 1, \beta = 0$ ) and the backward scheme ( $\alpha = 0, \beta = 1$ ),

$$R_{\text{forward}} = R_{\text{backward}} = \frac{\arctan \kappa \Delta t}{\kappa \Delta t}$$

- Using the Taylor expansion of arctan:

$$\arctan x = x - \frac{x^3}{3} + \dots$$

it follows that for good numerical resolution ( $\kappa \Delta t \ll 1$ ),

$$R_{\text{forward}} = R_{\text{backward}} \approx 1 - \frac{(\kappa \Delta t)^2}{3}$$

So the forward scheme and the backward scheme are decelerating.

- For the trapezium scheme ( $\alpha = \beta = 1/2$ ),

$$\begin{aligned} R_{\text{trapezoidal}} &= \frac{1}{\kappa \Delta t} \arctan \left( \frac{\kappa \Delta t}{1 - \frac{\kappa^2 \Delta t^2}{4}} \right) \\ &\approx 1 - \frac{(\kappa \Delta t)^2}{12} \end{aligned}$$

So the trapezium scheme is also decelerating, but the phase error is less than with the forward and the backward schemes.



Exercise: section 1 of the Jupyter notebook `time_discretization.ipynb`

## Important remark:

For the backward and the trapezoidal scheme,  $F(\phi^{n+1})$  is needed to compute  $\phi^{n+1}$ :

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \alpha F(\phi^n) + \beta F(\phi^{n+1})$$

Such schemes are called *implicit* schemes. Schemes for which only  $\phi^n$  is needed, are called *explicit* schemes.

For the oscillation equation  $F(\psi) = i\kappa\psi$ , so the solution is trivial. For other expressions for  $F(\psi)$ , this will not be the case!

## Important remark:

For the backward and the trapezoidal scheme,  $F(\phi^{n+1})$  is needed to compute  $\phi^{n+1}$ :

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \alpha F(\phi^n) + \beta F(\phi^{n+1})$$

Such schemes are called *implicit* schemes. Schemes for which only  $\phi^n$  is needed, are called *explicit* schemes.

For the oscillation equation  $F(\psi) = i\kappa\psi$ , so the solution is trivial. For other expressions for  $F(\psi)$ , this will not be the case!

**Implicit schemes are much more stable, but the cost-per-timestep is higher.**

Some examples of challenges for implicit schemes:

- nonlinear systems, e.g.  $F(\psi) = \sin(\psi)$

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \frac{1}{2} \sin(\phi^n) + \frac{1}{2} \sin(\phi^{n+1})$$

This usually has to be solved iteratively.

Some examples of challenges for implicit schemes:

- nonlinear systems
- systems with multiple variables become *coupled*, e.g. shallow water equations with 2 variables  $u$  and  $h$  (cfr. Lesson 5):

$$\begin{aligned}\frac{u^{n+1} - u^n}{\Delta t} &= -g \frac{\partial}{\partial x} (h^n + h^{n+1}) \\ \frac{h^{n+1} - h^n}{\Delta t} &= -H \frac{\partial}{\partial x} (u^n + u^{n+1})\end{aligned}$$

A linear system has to be solved.

Some examples of challenges for implicit schemes:

- nonlinear systems
- systems with multiple variables become *coupled*
- for systems involving spatial derivatives, the gridpoints become coupled, e.g. advection equation:

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} = -\frac{c}{2} \left( \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} + \frac{\phi_{j+1}^{n+1} - \phi_{j-1}^{n+1}}{2\Delta x} \right)$$

A (LARGE!) tridiagonal system has to be solved.

Multistage methods evaluate the RHS  $F(\phi)$  several times instead of one single time.

Example: Heun scheme (also called Runge-Kutta-2) which replaces  $\phi^{n+1}$  in the RHS of the trapezium scheme by a forward estimate:

$$\tilde{\phi}^{n+1} = \phi^n + \Delta t F(\phi^n) \quad \text{forward estimate of } \phi^{n+1}$$

$$\phi^{n+1} = \phi^n + \frac{\Delta t}{2} \left[ F(\phi^n) + F(\tilde{\phi}^{n+1}) \right] \quad \text{trapezium-like timestep}$$

Multistage methods evaluate the RHS  $F(\phi)$  several times instead of one single time.

Example: Heun scheme (also called Runge-Kutta-2) which replaces  $\phi^{n+1}$  in the RHS of the trapezium scheme by a forward estimate:

$$\tilde{\phi}^{n+1} = \phi^n + \Delta t F(\phi^n) \quad \text{forward estimate of } \phi^{n+1}$$

$$\phi^{n+1} = \phi^n + \frac{\Delta t}{2} \left[ F(\phi^n) + F(\tilde{\phi}^{n+1}) \right] \quad \text{trapezium-like timestep}$$

**Exercise:** what is the amplification factor of the Heun scheme applied to the oscillation equation  $F(\psi) = i\kappa\psi$ ?



Substituting  $F(\psi) = i\kappa\psi$  in the Heun scheme gives:

$$\begin{aligned}\tilde{\phi} &= (1 + i\kappa\Delta t)\phi^n \\ \phi^{n+1} &= \phi^n + \frac{i\kappa\Delta t}{2} (\phi^n + \tilde{\phi})\end{aligned}$$

Substituting  $F(\psi) = i\kappa\psi$  in the Heun scheme gives:

$$\begin{aligned}\tilde{\phi} &= (1 + i\kappa\Delta t)\phi^n \\ \phi^{n+1} &= \phi^n + \frac{i\kappa\Delta t}{2} (\phi^n + \tilde{\phi})\end{aligned}$$

Substituting  $\tilde{\phi}$  in the second equation gives

$$\phi^{n+1} = \left(1 + i\kappa\Delta t - \frac{(\kappa\Delta t)^2}{2}\right) \phi^n$$

Substituting  $F(\psi) = i\kappa\psi$  in the Heun scheme gives:

$$\begin{aligned}\tilde{\phi} &= (1 + i\kappa\Delta t)\phi^n \\ \phi^{n+1} &= \phi^n + \frac{i\kappa\Delta t}{2} (\phi^n + \tilde{\phi})\end{aligned}$$

Substituting  $\tilde{\phi}$  in the second equation gives

$$\phi^{n+1} = \left(1 + i\kappa\Delta t - \frac{(\kappa\Delta t)^2}{2}\right) \phi^n$$

So the amplification factor is

$$A = \frac{\phi^{n+1}}{\phi^n} = 1 + i\kappa\Delta t - \frac{1}{2} (\kappa\Delta t)^2$$

Substituting  $F(\psi) = i\kappa\psi$  in the Heun scheme gives:

$$\begin{aligned}\tilde{\phi} &= (1 + i\kappa\Delta t)\phi^n \\ \phi^{n+1} &= \phi^n + \frac{i\kappa\Delta t}{2} (\phi^n + \tilde{\phi})\end{aligned}$$

Substituting  $\tilde{\phi}$  in the second equation gives

$$\phi^{n+1} = \left(1 + i\kappa\Delta t - \frac{(\kappa\Delta t)^2}{2}\right) \phi^n$$

So the amplification factor is

$$A = \frac{\phi^{n+1}}{\phi^n} = 1 + i\kappa\Delta t - \frac{1}{2} (\kappa\Delta t)^2$$

which is second-order accurate (compare with Taylor expansion of exact amplification factor).

The modulus is

$$|A|^2 = 1 + \frac{1}{4} (\kappa \Delta t)^4 > 1$$

So the Heun scheme is unconditionally unstable!

The Matsuno scheme combines a forward estimate with a backward step:

$$\tilde{\phi}^{n+1} = \phi^n + \Delta t F(\phi^n) \quad \text{forward estimate of } \phi^{n+1}$$

$$\phi^{n+1} = \phi^n + \Delta t F(\tilde{\phi}^{n+1}) \quad \text{backward-like timestep}$$

The amplification factor is

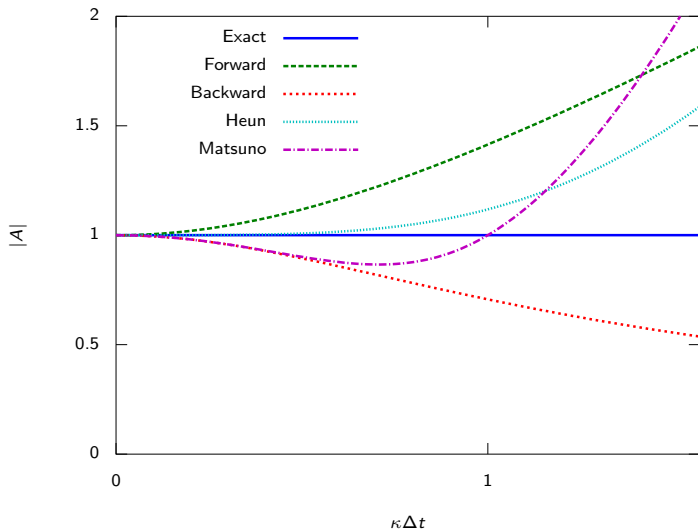
$$A = 1 + i\kappa\Delta t - (\kappa\Delta t)^2$$

which is first-order accurate.

The modulus is

$$|A|^2 = 1 - (\kappa\Delta t)^2 + (\kappa\Delta t)^4$$

So the Matsuno scheme is conditionally stable:  $\kappa\Delta t < 1$



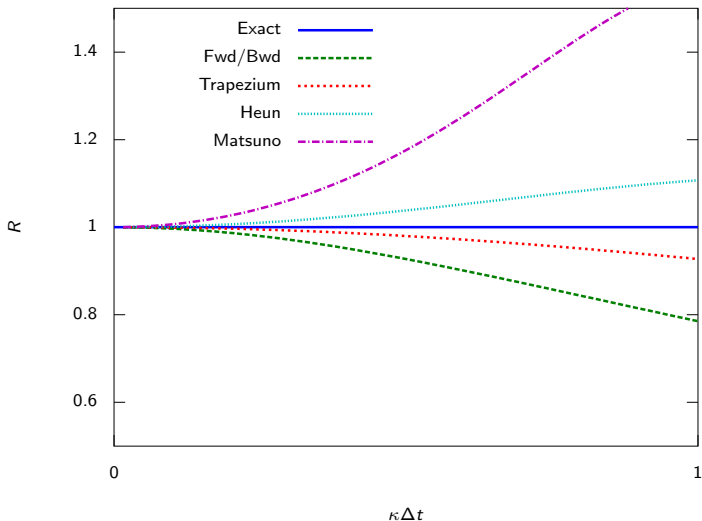
Approximated for  $\kappa\Delta t \ll 1$ ,

$$R_{\text{Heun}} \approx 1 + \frac{1}{6} (\kappa\Delta t)^2$$
$$R_{\text{Matsuno}} \approx 1 + \frac{2}{3} (\kappa\Delta t)^2$$

So the Heun scheme has less phase change than the Matsuno scheme.



## Multistage methods: phase change



The following scheme is quite popular because it's 4th order accurate:

$$q_1 = \Delta t F(\phi^n)$$

$$q_2 = \Delta t F(\phi^n + q_1/2)$$

$$q_3 = \Delta t F(\phi^n + q_2/2)$$

$$q_4 = \Delta t F(\phi^n + q_3)$$

$$\phi^{n+1} = \phi^n + \frac{1}{6} (q_1 + 2q_2 + 2q_3 + q_4)$$

This scheme is conditionally stable ( $|\kappa \Delta t| < 2.828$ ).

The drawback of multistage methods is that the right hand side  $F(\psi)$  needs to be evaluated several times. This makes them suitable for toy-models, but not for operational 3D NWP models.

Exercise: section 2 of the Jupyter notebook `time_discretization.ipynb`

It is possible to reuse information from the previous timestep  $\phi^{n-1}$ :

$$\phi^{n+1} = \alpha_1 \phi^n + \alpha_2 \phi^{n-1} + \beta_1 \Delta t F(\phi^n) + \beta_2 \Delta t F(\phi^{n-1})$$

Such a scheme will be at least second-order if

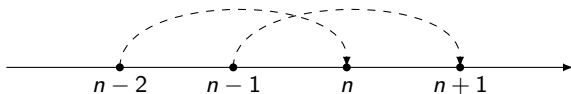
$$\alpha_1 = 1 - \alpha_2 \quad \beta_1 = \frac{1}{2}(\alpha_2 + 3) \quad \beta_2 = \frac{1}{2}(\alpha_2 - 1)$$

We limit ourselves to the following (second order) schemes:

|                 | $\alpha_1$ | $\alpha_2$ | $\beta_1$     | $\beta_2$      |
|-----------------|------------|------------|---------------|----------------|
| Leapfrog        | 0          | 1          | 2             | 0              |
| Adams-Bashforth | 1          | 0          | $\frac{3}{2}$ | $-\frac{1}{2}$ |

The leapfrog scheme for the oscillation equation:

$$\phi^{n+1} = \phi^{n-1} + 2i\kappa\Delta t\phi^n$$



The amplification factor satisfies

$$A^2 - 2i\kappa\Delta tA - 1 = 0$$

so

$$A_{\pm} = i\kappa\Delta t \pm \sqrt{1 - \kappa^2\Delta t^2}$$

which is second-order accurate.

There are 2 solutions corresponding to 2 'modes'. If  $\kappa\Delta t \rightarrow 0$  then  $A_+ \rightarrow 1$  and  $A_- \rightarrow -1$ . Therefore,

$A_+ \sim$  physical mode

$A_- \sim$  purely artificial 'computational' mode

- If  $|\kappa\Delta t| \leq 1$ ,  $|A_+| = |A_-| = 1$  so the scheme is stable.
- If  $\kappa\Delta t > 1$

$$|A_+| = \left| i\kappa\Delta t + i\sqrt{(\kappa^2\Delta t^2 - 1)} \right| > |i\kappa\Delta t| > 1$$

So the scheme is unstable.

- If  $\kappa\Delta t < -1$  then  $|A_-| > 1$  so the computational mode is unstable.

The computational mode is easy to analyse in the case  $\kappa = 0$ :

$$\phi^{n+1} = \phi^{n-1}$$

and the roots are  $A_+ = 1, A_- = -1$

For the initial condition  $\phi^0 = C$ :

$$\phi^2 = \phi^4 = \phi^6 = \dots = C$$

$\phi^1$  is usually determined by another two-time-level scheme:  $\phi^1 = C + \epsilon$  with  $\epsilon$  some error. Then

$$\phi(t) = \{ C, C + \epsilon, C, C + \epsilon, C, C + \epsilon \dots \}$$

So the error  $\epsilon$  stays in the solution.

The relative phase change of the physical mode of the leapfrog scheme is given by

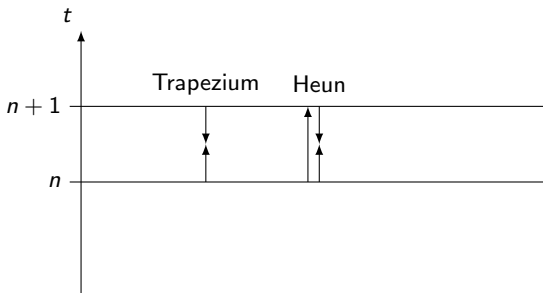
$$R_{\text{leapfrog}} = \frac{1}{\kappa \Delta t} \arctan \left( \frac{\kappa \Delta t}{\sqrt{1 - \kappa^2 \Delta t^2}} \right) \\ \approx 1 + \frac{1}{6} (\kappa \Delta t)^2$$

So the leapfrog scheme is accelerating.



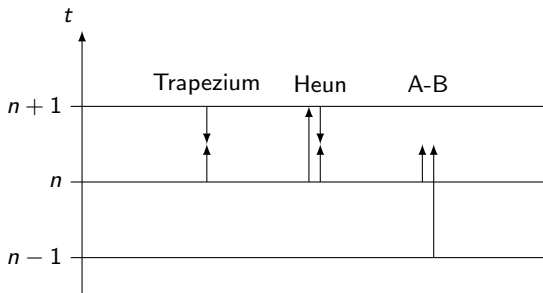
The trapezium scheme is very attractive (neutral, 2nd order, small phase error), but it is implicit: it requires  $F(\phi^{n+1})$  to determine  $\phi^{n+1}$ .

The Heun scheme attempts to approximate the trapezium scheme by using a forward estimate in the RHS:



The trapezium scheme is very attractive (neutral, 2nd order, small phase error), but it is implicit: it requires  $F(\phi^{n+1})$  to determine  $\phi^{n+1}$ .

The Heun scheme attempts to approximate the trapezium scheme by using a forward estimate in the RHS:



The Adams-Bashforth scheme estimates  $F\left(\phi^{n+\frac{1}{2}}\right)$  by extrapolating from  $F(\phi^n)$  and  $F(\phi^{n-1})$ .

The Adams-Bashforth scheme writes:

$$F\left(\phi^{n+\frac{1}{2}}\right) = \frac{3}{2}F\left(\phi^n\right) - \frac{1}{2}F\left(\phi^{n-1}\right)$$

$$\phi^{n+1} = \phi^n + \Delta t F\left(\phi^{n+\frac{1}{2}}\right)$$

For the oscillation equation, the amplification factor satisfies

$$A^2 - \left(1 + \frac{3}{2}i\kappa\Delta t\right)A + \frac{1}{2}i\kappa\Delta t = 0$$

with two modes,

$$A_{\pm} = \frac{1}{2} \left[ 1 + \frac{3}{2}i\kappa\Delta t \pm \sqrt{1 - \frac{9}{4}(\kappa\Delta t)^2 + i\kappa\Delta t} \right]$$

For good temporal resolution  $\kappa\Delta \leq 1$ ,

$$|A_+|_{A-B2} \approx 1 + \frac{1}{4} (\kappa\Delta t)^4$$

$$|A_-|_{A-B2} \approx \frac{1}{2} \kappa\Delta t$$

$$R_{A-B2} \approx 1 + \frac{5}{12} (\kappa\Delta t)^2$$

In NWP application we want strict stability, so this weak instability makes Adams-Bashforth unattractive.

In contrast, the leapfrog mode contains the artificial computational mode. However, this mode can be controlled by a Robert-Asselin filter.

We modify the leapfrog-scheme by adding a filter step:

$$\phi^{n+1} = \overline{\phi^{n-1}} + 2\Delta t F(\phi^n)$$

normal leapfrog

$$\overline{\phi^n} = \phi^n + \gamma \left( \overline{\phi^{n-1}} - 2\phi^n + \phi^{n+1} \right)$$

apply filter

where typically  $\gamma = 0.06$  in meteorological models.

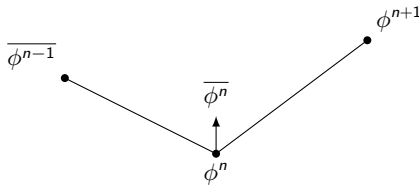
We modify the leapfrog-scheme by adding a filter step:

$$\phi^{n+1} = \overline{\phi^{n-1}} + 2\Delta t F(\phi^n) \quad \text{normal leapfrog}$$

$$\overline{\phi^n} = \phi^n + \gamma \left( \overline{\phi^{n-1}} - 2\phi^n + \phi^{n+1} \right) \quad \text{apply filter}$$

where typically  $\gamma = 0.06$  in meteorological models.

The term  $\overline{\phi^{n-1}} - 2\phi^n + \phi^{n+1}$  can be interpreted as a temporal filter that damps the high frequencies (esp. the  $2\Delta t$ -mode): if  $\overline{\phi^{n-1}} > \phi^n$  and  $\phi^{n+1} > \phi^n$ , then  $\overline{\phi^n}$  will be increased:



Amplification of Asselin-leapfrog:

$$A_{\pm} = \gamma + i\kappa\Delta t \pm \sqrt{(1 - \gamma)^2 - \kappa^2\Delta t^2}$$

For small  $\kappa\Delta t$ ,

$$A_+ \approx 1 + i\kappa\Delta t - \frac{(\kappa\Delta t)^2}{2(1 - \gamma)}$$

$$A_- \approx -1 + 2\gamma + i\kappa\Delta t + \frac{(\kappa\Delta t)^2}{2(1 - \gamma)}$$

while expansion of the exact solution is:

$$A_e = e^{i\kappa\Delta t} = 1 + i\kappa\Delta t - \frac{1}{2}(\kappa\Delta t)^2 - i\frac{1}{6}(\kappa\Delta t)^3 + O[(\kappa\Delta t)^4]$$

So the Robert-Asselin filter degrades the accuracy of the scheme from 2nd order to 1st order.

The modulus of the amplification factor becomes:

$$|A_+| \approx 1 - \frac{\gamma}{2(1-\gamma)}(\kappa\Delta t)^2$$
$$|A_-| \approx (1-2\gamma) + \frac{\gamma}{2-6\gamma+4\gamma^2}(\kappa\Delta t)^2$$

Note that the computational mode is (slightly) damped.

The relative phase change becomes:

$$R_+ \approx 1 + \frac{1+2\gamma}{6(1-\gamma)}(\kappa\Delta t)^2$$



Exercise: section 3 of the Jupyter notebook `time_discretization.ipynb`

- We work on the *oscillation equation*, to focus on the time differencing aspect
- A scheme is characterized by
  - ▶ amplitude: amplifying, neutral or damping
  - ▶ relative phase change: accelerating or decelerating
  - ▶ explicitness or implicitness
  - ▶ order of accuracy
- Concrete examples:
  - ▶ Two time-level schemes: forward, backward, **trapezium**
  - ▶ Multistage schemes: Heun, Matsuno, **Runge-Kutta-4**
  - ▶ Three time-level schemes: **leapfrog** and Adams-Bashforth  
⇒ computational mode, filter
- There is no 'best' scheme: every scheme has disadvantages.
- What matters is methodology!